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SOFTWARE

Instrumented systems

Crawlers, pipelines, search indexes, reporting layers. Built around the actual data model: queryable; auditable; extendable.

Automation · Search · Data Pipelines · CLI · Reporting

Workflow automation

Manual processes converted to defined pipelines: structured inputs, quality gates, clean handoffs.

Automation · Search · Data Pipelines · CLI · Reporting

SYSTEMS

Architecture before components

Topology, constraints, and failure modes defined first.

Topology · Process Design · Constraints · Documentation · Operations

Process formalization

Informal rules surfaced, named, and documented.

Topology · Process Design · Constraints · Documentation · Operations

DESIGN

DC lighting architecture

Distribution layout, driver selection, control protocol, dimming behavior, serviceability.

Low Voltage · DC Distribution · Typst · Specification · Diagrams

Technical publication

Specifications, diagrams, and installation logic precise enough to build from. (Clear enough to hand to a non-specialist without a walkthrough.)

Low Voltage · DC Distribution · Typst · Specification · Diagrams

RESEARCH

Primary source analysis

Datasheets, standards, supplier catalogs, field constraints.

Standards · Supply Chain · Specs · Field Constraints · Technical Writing

Actionable synthesis

Gap between specification and real installation conditions: identified, closed, documented.

Standards · Supply Chain · Specs · Field Constraints · Technical Writing

IMPLEMENTATION

Full-cycle execution

Layout, sourcing, commissioning, verification – without coordination loss from splitting phases across specialists who don't share a system model.

Installation · Commissioning · Prototyping · Verification · Low Voltage

Constraint-driven prototyping

Early builds targeting the hard problem: thermal behavior, dimming continuity, distribution loss, control latency. Constraint solved before the system is committed.

Installation · Commissioning · Prototyping · Verification · Low Voltage

ENGAGEMENT

Where this applies

Architecture undecided. Requirements rough. Existing system producing inconsistent results with no clear cause. That's the entry point.

Systems Thinking · Ambiguity · Technical Depth · Documentation · Handoffs

What it produces

A system with a defensible logic – documented, sourceable, buildable, and maintainable by someone other than the original designer.

Systems Thinking · Ambiguity · Technical Depth · Documentation · Handoffs

Features

Milestone

2019

Top-percentile GitHub contributor

[Source](#) ¹

Highlight

Imager

Rust / Node image optimization algorithm using brute force heuristics

[Source](#) ³

source :		39.00M (4 images)
kraken.io :		24M
jpegmini.com :		16M
compression.ai :		8.90M
imager :		4.20M

Achievement

728+ ★

GitHub stars on most popular authored OSS tool

[Source](#) ²

Review

Academic

"You are such a profound writer and thinker. It has been my privilege to be your instructor of record. One day I will say, I had him in my English class. I have such high hopes for you! Go conquer your world! You're awesome."

– Dr. Jim Birrell
[Research Project](#) ⁴

Early Career History

pre 2015

Space Monkey

Brief High School Internship

Worked under Tom Metge

2016

Uplynk / Verizon Digital Media Services

Worked as a Jr. Developer on the QA team.

Team lead: Asiel Brumfield.

Left to work on encrypted images.

2015

Galileo Processing

Galileo Processing

Worked under Robert Raver.

2020

2020 – UVU CS Grader

Utah Valley University

CS Grader – Computer Science

Bianca Ruiz

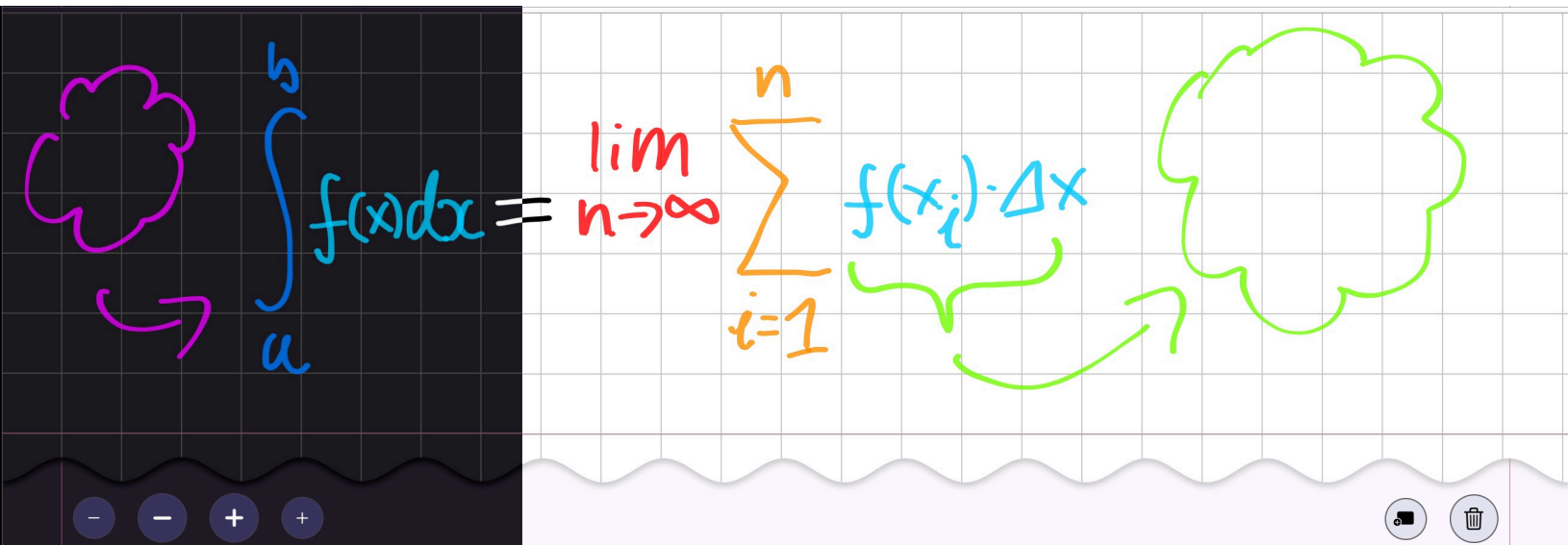
Overview links

1 <https://github.com/colbyn?tab=overview&from=2019-12-01&to=2019-12-31>

2 <https://github.com/imager-io/imager>

3 <https://github.com/imager-io/imager>

4 <https://github.com/colbyn/analog-computing-and-refs>



Spectrum Is the Real Light Source

THE HIDDEN LAYER

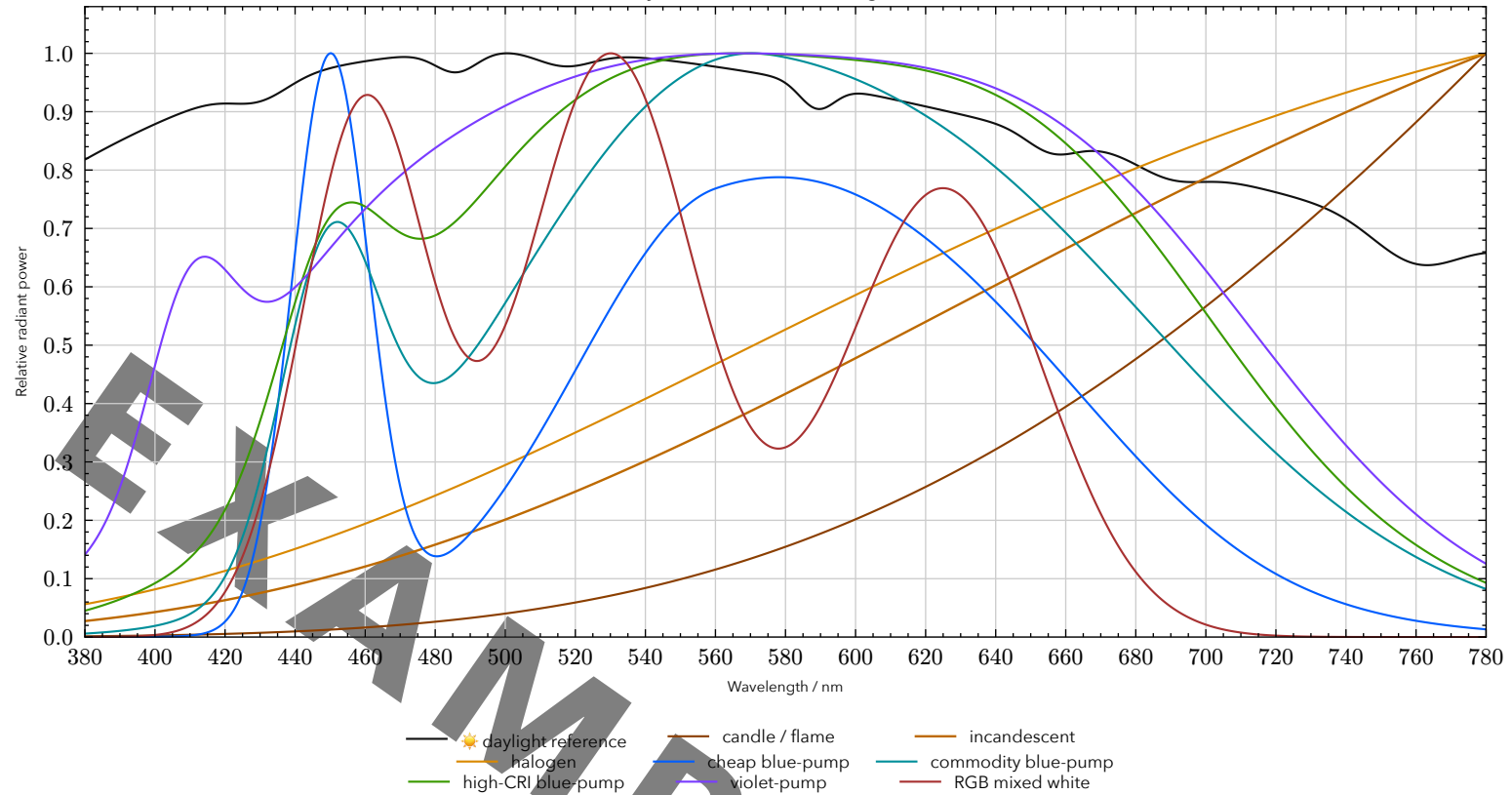
White light is not a substance.

It is a spectral construction that happens to land on a white appearance.

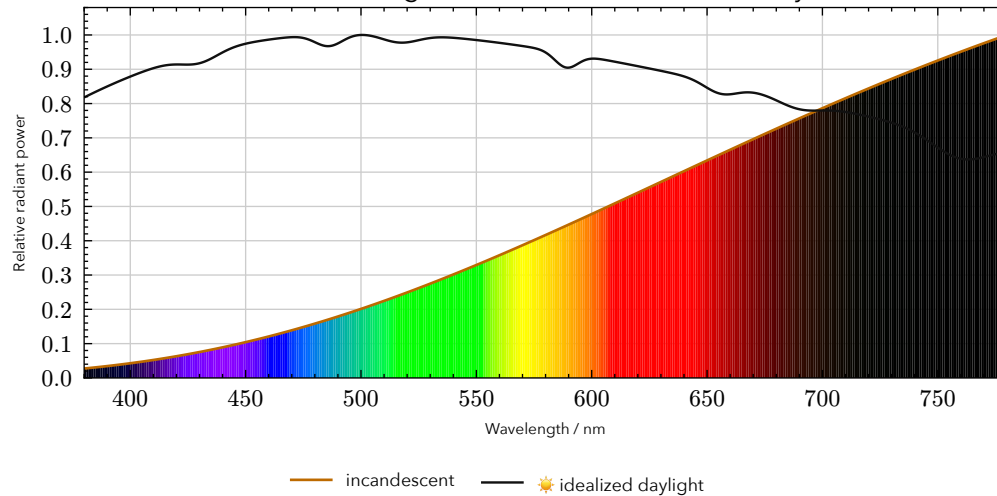
READING RULE

Do not ask only whether the light is warm or cool. Ask where the energy is, where it is missing, and how the spectrum is made.

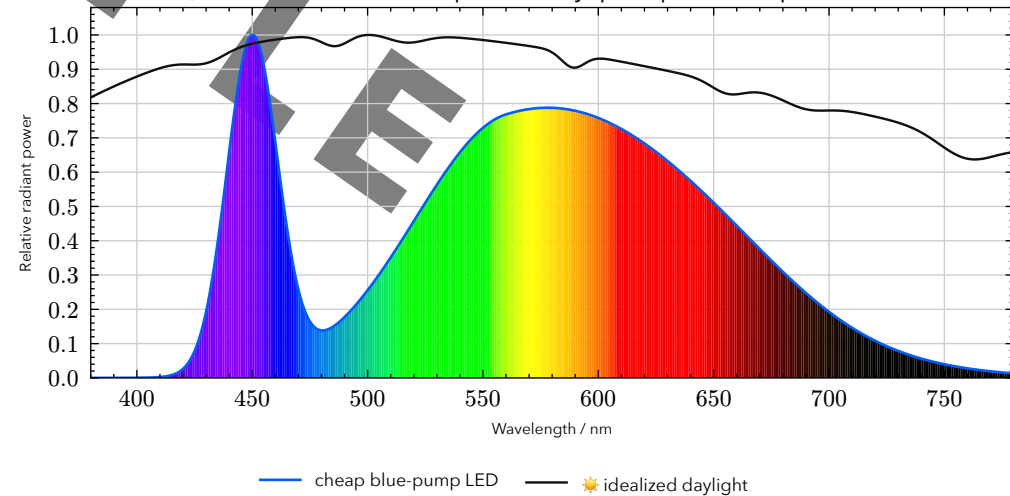
Different ways to make something that looks white



Thermal light: continuous, warm, red-heavy



Basic LED: spike, valley, phosphor hump



Incandescent is spectrally continuous, but tilted heavily toward long wavelengths. Its weakness is efficiency, not spectral assembly.

A basic blue-pump LED can look white while still having a hard pump spike, a cyan depression, and weak deep red.